

FUNDING THE CAPABILITY-RESOURCE GAP: A MODEL OF OPTIMAL SCIENCE FUNDING

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ABSTRACT. The grant funder’s challenge is to maximize the impact of a fixed budget under uncertainty about the researchers who might receive it. We develop a Bayesian model of this challenge: researchers differ in their epistemic capabilities and in resources; research output requires both, and funding compounds capability across rounds. We show that the optimal allocation funds the capability-resource gap: grants increase with capability and decrease with existing resources. Track records underdetermine the gap; peer review supplies an additional, noisy signal of capability. We find that this signal’s value increases with capability inequality and with its informativeness, and that under heavy-tailed capability distributions even a coarse signal suffices. We further find that uniform seed grants and lotteries, which forgo targeting, are cheap exactly where review is worth little, and that whom to fund matters far more than when to fund.

1. INTRODUCTION

Science funding is large and consequential. How best to allocate it remains poorly understood. The United States performed \$937 billion of R&D in 2023 and an estimated \$993 billion in 2024, roughly a fifth of it funded by the federal government (National Center for Science and Engineering Statistics 2026; National Science Board 2026). In spite of this, basic dimensions of allocation are contested. Whether funding should be concentrated on a few researchers or spread across many is a long-running dispute (Aagaard et al. 2020). Whether peer review is effective is disputed: among funded NIH grants, percentile scores predicted subsequent productivity barely better than chance (Fang et al. 2016), while other designs find modest but real predictive power (Gallo et al. 2014; Li and Agha 2015; Park et al. 2015). And whether review should be replaced, in part, by lotteries has moved from proposal to practice (Roumbanis 2019; Heyard et al. 2022). These disputes concern a single underlying problem: what a funder with a fixed budget and imperfect knowledge of researchers should do. This paper develops a

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model of that problem and works out its logic: whom to fund, whether to spend early or wait and learn, what information about researchers is worth buying, and what egalitarian alternatives cost.

Consider the challenge of the grant funder, whether a federal agency disbursing billions or a private foundation disbursing millions: you have a fixed budget and must choose how to allocate it across researchers and research projects to create the most positive impact. There are two intuitive schemes. Fund the track record: give to those who have produced the most, since past production is the best evidence of future production. Fund the under-resourced: give to those who have the least, since a dollar goes further where dollars are scarce. Each scheme can be intuitively compelling. Both, we will show, are suboptimal, and for the same reason: each accounts for only one of the two quantities the impact of a grant depends on.

We model this decision situation as follows. A funder with a fixed budget allocates grants, over multiple rounds, to researchers who differ in two ways: epistemic capability, what a researcher knows and can do, and resources, what a researcher has to work with. Research output requires both and is limited by the scarcer of the two; the funder can observe output, but not capability or resources themselves. A grant adds to a researcher's resources in the round it is given and, because doing research builds expertise, compounds their capability in later rounds. Each round, the funder observes the output that results, updates its beliefs about the researchers, and allocates again.

Our results are as follows. In the ideal condition where the funder has complete information regarding researchers' capabilities and resources, the optimal allocation funds the capability-resource gap: each funded researcher receives a grant increasing in their capability and decreasing in their existing resources, and the budget determines how many researchers receive funding at all. Each intuitive scheme fails because it accounts for only one of the two quantities. In reality the funder lacks direct access to the gap, and track records alone underdetermine it: we show that a funder relying on the track record performs far worse than one that estimates the gap.

Peer review supplies an additional signal of capability. Its value for a Bayesian funder is the expected value of its reduction of the funder's uncertainty regarding the gap, and that value is greatest where capability is unequally distributed and review informative; in fields with heavy-tailed distributions of capability, even a coarse signal captures most of this value. The same reasoning elucidates the cost of the egalitarian alternatives: uniform seed grants and lotteries cost the

value of targeting they displace, little where review is worth little, most exactly where review is worth most. The budget is a variable throughout, so our results speak across funder scales: a small foundation, whose grants are modest beside researchers' total resources, is the funder for whom targeting matters most per dollar. Whether optimal funding should concentrate on a few researchers or spread across many depends on the size of the funder's budget relative to researchers' total resources. We show how strategic timing of when one provides funds matters far less than strategic targeting of whom one provides funds: how much of the budget to spend early rather than late shifts with the funder's budget relative to researchers' total resources, but the difference is small relative to the difference between funding the right and the wrong researchers.

Two substantive assumptions bound these results. Grants are consumed, a grant's durable residue (skills, publications, standing) being carried as capability; and capability and resources are complements throughout, neither fully substituting for the other. We defend each assumption where it is used.

We proceed as follows. §2 situates our model in the literatures on funding allocation, peer review, and lotteries. §3 presents our model. §4 derives the optimal allocation. §5 shows why track records and small grants cannot reveal the gap. §6 measures the value of peer review and relates it to the empirical evidence on review. §7 examines the timing of funding, including the strongest argument for spending early. §8 prices seed grants and lotteries. §9 varies the production technology and revisits the concentration question. §10 discusses implications and limitations, and §11 concludes.

2. RELATED LITERATURE

Our contribution sits at the intersection of four literatures: the empirical debate over concentrating or dispersing funds, the evidence on the efficacy of peer review, the case for funding lotteries, and formal models of science funding.

Whether funders should concentrate resources on a few researchers or spread them across many is a long-running dispute, and concentration is the current practice: the top 1% of NIH principal investigators receive close to 10% of research project grant funding, a share that has risen since 1985 (Katz and Matter 2020; Lauer and Roychowdhury 2021). Empirical studies of funded portfolios generally find diminishing marginal returns to funding per researcher, and reviews of this literature conclude that the evidence favors more dispersal than current practice (Mongeon et al. 2016; Aagaard et al. 2020). Concentration is also self-reinforcing: early grant winners accumulate advantages beyond the grant itself (Bol et al. 2018), a dynamic that formal models show can reward questionable research practices over

a career (Heesen 2024). This dispute typically concerns the returns to funding: whether output per researcher increases or decreases as grants concentrate. In our model, the answer also depends on the size of the funder’s budget relative to researchers’ total resources: with a budget large enough to relieve most bottlenecks, the optimal allocation itself reaches nearly every researcher, so little is lost by spreading funds widely; with a tight budget, effectiveness requires concentrating funds on the researchers with the largest capability-resource gaps (§9). The value of any targeting, in turn, depends on what the funder can know about researchers (§6).

The efficacy of peer review is contested. Among funded NIH grants, percentile scores predict subsequent productivity barely better than chance (Fang et al. 2016), and reviewers assessing the same application agree only weakly (Pier et al. 2018). Analyses with different designs find real but modest predictive power: better scores predict more publications, citations, and patents, including under controls for investigator history and institution (Li and Agha 2015), and a funding natural experiment finds higher-scored proposals outproduce lower-scored ones (Park et al. 2015). Both sides of this dispute measure review’s accuracy: how well scores predict outcomes among funded applicants. Our model instead measures review’s value: the expected value of its reduction of the funder’s uncertainty regarding the capability-resource gap. That value depends on the field; the same noisy signal can be worth a great deal in one field and nothing in another (§6), and no single number settles the question.

Doubts about review have made lotteries a live proposal. Advocates argue that randomizing among fundable applications saves review costs, reduces bias, and honestly reflects review’s limited ability to discriminate among fundable proposals (Fang and Casadevall 2016; Avin 2019a; Roumbanis 2019); several funders have implemented partial lotteries: the Health Research Council of New Zealand has allocated its Explorer Grants by lottery since 2013 (Liu et al. 2020), the Swiss National Science Foundation randomizes among ties (Heyard et al. 2022), and a German funding line runs a lottery before review (Luebber et al. 2025). The existing case prices what a lottery saves: Gross and Bergstrom (2019) model grant competition as a contest in which scientists invest costly effort in proposals, show that at low paylines this effort can rival the scientific value of the research funded, and show that a partial lottery among proposals above a threshold decouples the waste from the payline. What remains unpriced is what a lottery forgoes. Our model supplies that half of the ledger: the value of targeting, which runs from negligible, where budgets are ample, signals uninformative, or capability evenly

spread, to substantial, where capability is heavy-tailed and budgets tight (§8). Together the two halves price the choice field by field.

Formal models of science funding have concentrated on the incentive costs of competition and on the value of exploration. Beyond the contest model above, Gross and Bergstrom (2025) find, in a game-theoretic model of contests for scarce rewards, that intensifying competition pushes scientists toward higher-risk, higher-return projects. Avin (2019b) simulates funding strategies on an epistemic landscape and finds that allocation with an explicitly random component significantly outperforms a peer-review-like strategy on large landscapes, because review concentrates effort on regions already known to be productive. Sorzano and Pueche-Granados (2026) treat allocation as a decision problem under heavy-tailed uncertainty about project value, pairing an empirical bibliometric signal with a biased-lottery mechanism. In each of these models, what funding buys is a single quantity, a project’s chance of success or a researcher’s productivity, and none treats the timing of spending as the funder’s choice. In our model, a researcher’s output increases in both capability and resources but is limited by the scarcer of the two, and the funder observes neither directly.

Our formulation and analysis of the capability-resource gap allows us to address questions raised by these literatures: how widely funds should be spread (§9), what peer review is worth (§6), and when lotteries are defensible (§8).

3. OUR MODEL

A funder faces a population of n researchers over T funding rounds. Researcher i has epistemic capability K_i and resources R_i . Initial capabilities and resources are drawn from Pareto distributions with tail parameters α_K and α_R ; smaller tail parameters produce more unequal fields, and a correlation parameter ρ allows capable researchers to tend to be well resourced. The funder knows these distributions; it never observes any researcher’s capability or resources directly.

In each round, researcher i produces research output¹ at random, in discrete units, with expected output²

$$\lambda_i = A \Lambda(K_i, R_i),$$

where A is a productivity constant and Λ is a production function that increases in both arguments but is limited by the scarcer of the two. Our main analysis

¹ We are agnostic about the interpretation of output units: they may be read as publications, as findings, or as units of scientific value.

² Output in each round is drawn from a Poisson distribution with rate λ_i , independently across researchers and rounds given the state.

takes Λ to be the harmonic mean:

$$\lambda_i = \frac{2AK_iR_i}{K_i + R_i}.$$

The harmonic mean makes capability and resources complements without making either an absolute requirement: a capable researcher with thin resources still produces, by working more slowly or borrowing equipment, but no surplus of one input fully substitutes for a shortage of the other. The harmonic mean sits between two boundary cases that fail in opposite directions: Cobb-Douglas, under which a capable researcher can compensate almost fully for vanishing resources, and Leontief, under which no compensation is possible at all. §9 repeats our analysis across the full family between these boundaries.

Capabilities increase through research. Between rounds,

$$K_i \leftarrow K_i + \epsilon \Lambda(K_i, R_i),$$

where ϵ is the compounding rate: doing research builds expertise, and the same bottleneck that limits output limits growth. Growth requires both inputs, and capability never declines. Resources, by contrast, do not accumulate: resources in a round are baseline plus grant, $R_i = R_{i0} + g_i$. The baseline R_{i0} recurs every round, representing the time and support a researcher has independent of the funder, while grants are consumed in the round they are given. This is a substantive assumption, not bookkeeping: whatever endures from a grant, the skills, publications, and standing it produces, is carried by the capability update, so funding's lasting effect flows through what researchers become rather than what they keep. Science is in this respect an accumulation process: success begets the means of further success.

The funder's total budget is fixed before any outcomes are observed. We parameterize it by a budget scale b : at $b = 0.5$, the purse equals the community's expected baseline resources for one round, and the purse scales linearly in b . (The exact normalization is in Appendix [X].) The budget does not grow with the horizon: a longer horizon varies the timing of spending, never its amount. Strategies that do not plan ahead spend an equal share of the purse each round. Forward-looking strategies may spend on any schedule: each round, they re-plan the entire remaining budget over the remaining horizon and execute the current round's allocation.

The funder learns as it goes. It observes each researcher's track record, the output they have produced to date, as it accumulates, and two further signals are available. A resource signal, observed once at the start, gives a noisy reading of each researcher's baseline resources (noise τ_R), as one might obtain from institutional information. A review signal, drawn afresh in each round that a strategy uses it,

Strategy	Information	Planning
No funding	none	none
Uniform	none	none
Track record	track record	none
Myopic, without review	track record + resource signal	current round
Forward, without review	track record + resource signal	remaining horizon
Myopic, with review	track record + resource signal + review	current round
Forward, with review	track record + resource signal + review	remaining horizon

TABLE 1. The funding strategies compared.

gives a noisy reading of capability (noise τ_K), as one might obtain from proposal evaluation and expert judgment. The review signal’s informativeness is governed by τ_K : the smaller the noise, the more informative the review. Bayesian strategies update beliefs about every researcher after each round and allocate on posterior expected marginal returns.

We compare seven funding strategies, which differ along two dimensions: information, with or without the review signal, and planning, myopic or forward-looking. A myopic strategy optimizes expected output in the current round; a forward-looking strategy optimizes expected output over the whole remaining horizon. Three non-Bayesian baselines complete the set: no funding, against which every result is normalized; uniform funding, which splits each round’s tranche equally; and track-record funding, which allocates in proportion to the previous round’s output. Uniform floors and lotteries, which modify these strategies rather than add to them, are introduced in §8.

Our conclusions rest either on analytic results supported by proofs or on simulation results that are established across the following parameter ranges.³ The tail parameters α_K and α_R range from 1.3 to 3.5. Distributions of research productivity are heavy-tailed: in Lotka’s classic data, the number of authors producing x papers falls off as x^{-2} , and later studies estimate exponents mostly between 2 and 3 (Lotka 1926; Nicholls 1989), corresponding to Pareto tail parameters near 1 to 2; our range covers this territory and extends to markedly more equal fields. The budget scale b ranges from 0.1 to 1, from a funder whose purse is small beside the community’s own resources to one able to relieve most researchers’ bottlenecks; §5 and §7 also examine deeper purses. The compounding rate ϵ ranges from near zero, where funding buys only current output, to 0.85, where this round’s funding substantially raises later capability. The review noise τ_K ranges from 0.05, nearly perfect review, to 20, nearly uninformative review, a range wide enough to contain

³ Unless a figure states otherwise, unstated parameters take the values $T = 2$, $n = 50$, $\epsilon = 0.1$, $b = 0.5$, $\alpha_K = \alpha_R = 2$, $\tau_K = \tau_R = 1$, and $\rho = 0$.

the noise implied by empirical estimates of review’s predictive accuracy (§6). The correlation ρ ranges from -0.5 to 0.8 , and the horizon T from one round to ten.

Randomness enters our model through the initial population draws, the signals, and realized output. Realized output shapes the funder’s beliefs and, through them, its allocations. Capability growth, by contrast, follows expected output: the update above uses Λ itself, so a researcher’s accumulation depends on their capability and resources, not on luck in any given round. This is a modeling choice: it treats a round as containing enough separate acts of research for luck to average out of accumulation, while leaving the funder’s inference fully exposed to noise. For the same reason, we score each strategy by the sum of expected outputs along its run, which equals realized output averaged over repetitions of the same allocations; this removes simulation noise from comparisons without favoring any strategy. Each configuration runs on 200 independently drawn populations, with common random numbers across strategies, so every comparison between strategies is paired. Throughout, we report each strategy’s improvement over no funding, in percent. The full specification, parameter table, and code are in Appendix [X].

With the model in place, the funder’s problem is exact: choose grants, given beliefs, to maximize expected total output over the horizon. §4 solves the case in which the funder has complete information regarding researchers’ capabilities and resources.

4. THE OPTIMAL ALLOCATION: FUND THE CAPABILITY-RESOURCE GAP

First, consider the simple case where the funder has complete information regarding researchers’ capabilities and resources. Further, we initially restrict our attention to a single round of funding, temporarily putting aside the contribution of resources to the accumulation of researchers’ capabilities across rounds. The funder’s problem is to divide the round’s tranche among researchers so as to maximize expected output:

$$\max_{g_1, \dots, g_n \geq 0} \sum_i A \Lambda(K_i, R_{i0} + g_i) \quad \text{subject to} \quad \sum_i g_i \leq \text{the round's tranche.}$$

Now, consider the marginal value of a dollar for such a funder. Under the harmonic production function,

$$\frac{\partial \lambda_i}{\partial g_i} = \frac{2A K_i^2}{(K_i + R_{i0} + g_i)^2}.$$

The marginal value of funding researcher i increases with their capability and decreases with the resources they already hold, including grants. An allocation is optimal exactly when no dollar can be moved to a researcher who can produce

more value with it, that is, when marginal values are equal across all funded researchers and no unfunded researcher’s marginal value exceeds theirs.

Proposition 1 (the gap rule). *For every researcher, the optimal grant fills the gap, if any, between the resources they hold and a target proportional to their capability:*

$$g_i^* = \max(c K_i - R_{i0}, 0),$$

where the constant $c > 0$ is set by spending the budget.⁴

We can understand the rule as follows. Each researcher’s target is c times their capability, and their gap is the difference between their target and the resources they hold. The rule grants every researcher with a positive gap exactly their gap, and grants nothing to the rest; the constant c is defined as the value at which these grants sum to the budget. Because c is defined by the budget, the budget always covers the gaps: a question of which gaps to fund first does not arise. When the budget is small, c is small: targets are low, few researchers have positive gaps, and their grants are small. When the budget is large, c is large: targets are high, and funding extends across nearly the whole population. In both cases the funded researchers, those with positive gaps ($R_{i0} < c K_i$), are those with the least resources relative to their capability, and these are the researchers for whom a dollar has the greatest marginal value.⁵ The condition $R_{i0} < c K_i$ draws a frontier through the population, the funding frontier, separating the funded from the unfunded (Figure 1).

The gap rule explains why each of the intuitive funding schemes discussed earlier fails. Funding the track record fails because output reflects resources as well as capability: a productive researcher whose resources approach or exceed their target has a small or negative gap, and the marginal dollar buys little there. The failure is not merely a shortfall from the optimum: because track-record funding channels money toward researchers whose bottleneck is not resources, it can produce less output than spreading the budget evenly, that is, than not targeting at all.⁶ Funding the under-resourced fails for the mirror-image reason: scarcity is no evidence of capability, and dollars allocated to researchers of low capability buy little however scarce their resources. It, too, can produce less than not targeting

⁴ $c = \sqrt{2A/\nu} - 1$, with ν the common marginal value at the optimum. The proof is in Appendix [X].

⁵ The optimality condition behind the rule, funding to a common marginal value, is the water-filling solution familiar from information theory (Cover and Thomas 2006). If all researchers shared one capability, the rule would reduce to filling each researcher’s gap below one common target, the poverty-gap transfer of development economics (Foster et al. 1984); our rule scales the target with capability.

⁶ Appendix [X] gives a two-researcher example.

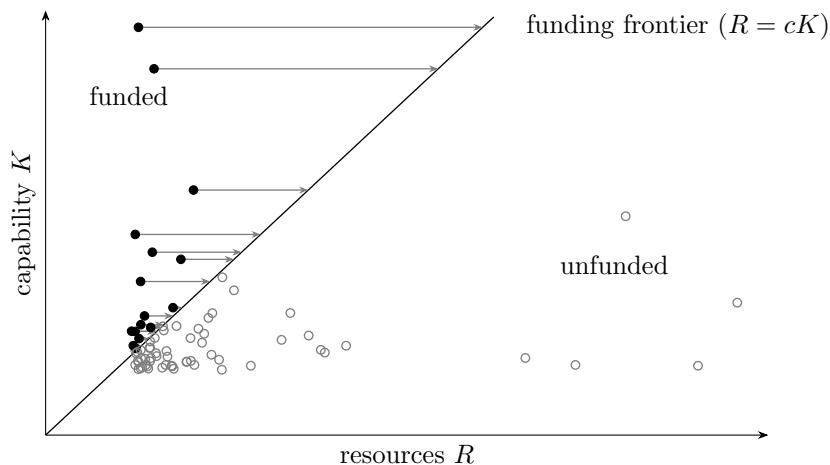


FIGURE 1. The gap rule in one population. Each point is a researcher, positioned by their resources and capability. The line is the funding frontier, $R = cK$, here at budget scale $b = 0.1$. The rule funds every researcher whose resources fall below the frontier (filled points), and each grant, drawn as an arrow, moves its researcher exactly to the frontier: the arrow's length is the researcher's gap, $cK_i - R_{i0}$. Researchers at or beyond the frontier (open points) receive nothing.

at all.⁷ Each scheme accounts for one of capability and resources, whereas the marginal value of a dollar is determined by how a researcher's resources compare with their capability, and so responds to both.

How much targeting matters depends on the budget. A researcher's expected output increases with every dollar of resources, but with diminishing marginal returns, and it never exceeds $2AK_i$: expected output is bounded above by a quantity set by capability, which it approaches but does not attain. When the budget is large, optimal and uniform funding therefore differ little in the output they produce: under either, every researcher's output is near its upper bound, and the difference between the two vanishes as the budget increases (Appendix [X]). When the budget is small, dollars are scarce, researchers differ in the marginal value of a dollar, and output depends on where each dollar goes. Targeting matters where money is tight and little where it is ample (Figure 2). This comparative static recurs: it sets the price of overriding the optimal allocation with floors and lotteries (§8), and it determines when optimal funding concentrates rather than spreads (§9).

⁷ Again, Appendix [X] gives an example.

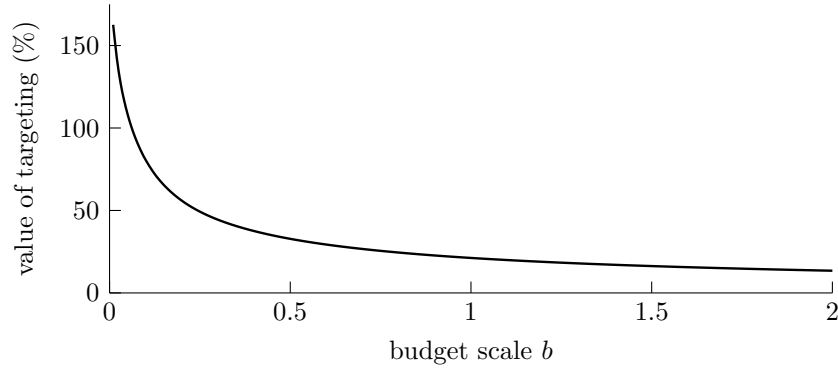


FIGURE 2. The value of targeting decreases in the budget. The value of targeting is the additional expected output from the optimal allocation over uniform funding, as a fraction of uniform funding's gain over no funding; the budget scale b is the budget as a fraction of the population's total baseline resources. The curve shows one population of $n = 75$ researchers drawn from the model's default distributions ($\alpha_K = \alpha_R = 2$, $\rho = 0$, $A = 1$); the value of targeting decreases toward zero as the budget increases (Corollary 2).

To use the gap rule, however, a funder requires an estimate of the gap between researchers' capabilities and their resources. We turn now to how such estimation can work: from the track record alone (§5), and with peer review added (§6).

5. WHAT THE TRACK RECORD CAN AND CANNOT SUPPLY

The funder observes track records: each researcher's realized output, round by round. A track record is evidence about the capability-resource gap, and the question for this section is how far that evidence goes.

Consider first what output can identify in principle. Expected output is one number, produced by capability and resources jointly: every pair of capability and resources with $2AKR/(K + R) = \lambda$ produces the same expected output λ . A researcher of high capability and modest resources and a researcher of modest capability and abundant resources can therefore produce identical records. Output does carry some information about capability, a lower bound: since expected output never exceeds $2AK$, a researcher who produces λ has capability above $\lambda/2A$. Beyond that bound, the record cannot separate the two researchers above, and these are exactly the researchers the gap rule treats most differently: the first may have the largest gap in the population, the second may have none. Cross-sectional output, however carefully recorded, underdetermines what the funder most needs to know.

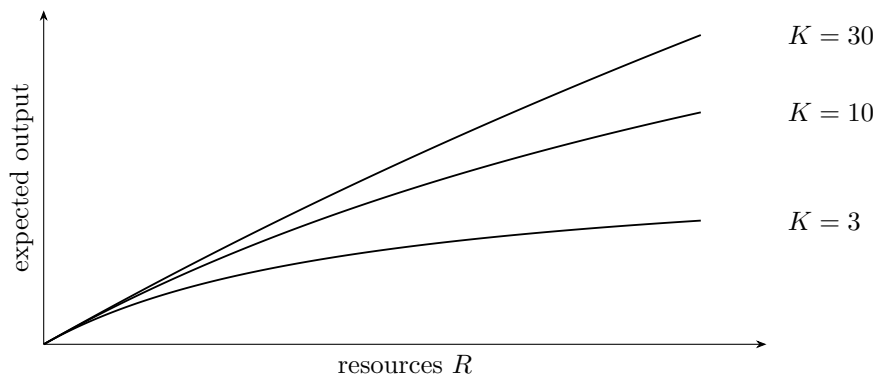


FIGURE 3. Thin grants are uninformative about capability. Expected output as a function of resources, for researchers of capability $K = 3, 10$, and 30 ($A = 1$). Every curve has slope $2A$ at the origin: where resources are small relative to capability, the curves coincide, and researchers of very different capabilities produce nearly identical output. The curves separate where resources are comparable to capability: observing output can identify capability only at depth.

Our model measures what this underdetermination costs. A Bayesian funder relying on records alone allocates far from the gap rule and remains far: its allocations stay only weakly correlated with the optimal grants, round after round, and its output falls well short of the complete-information benchmark.⁸ This shortfall is the span from records alone to complete information. §6 asks how much of that span a review signal recovers, and there we display this records-only baseline beside review signals of increasing informativeness.

The record is least informative exactly where the gap is largest. When a researcher's resources are small relative to their capability, expected output is approximately $2AR$: proportional to resources, nearly independent of capability (Figure 3). On a thin grant, a researcher of capability 1 and a researcher of capability 100 produce nearly indistinguishable output. Small grants therefore buy almost no information about who is capable; separating capability from resources by observing output requires funding depth, grants comparable to capability itself. In our simulations of a resource-poor community with no review signal, a funder allocating at standard depth gains almost nothing over uniform funding, while the same funder with deep grants recovers most of the value of discrimination (§7 measures this and its consequences for timing). For nascent and resource-poor fields the implication is direct: thin grants cannot reveal who is capable.

⁸ At the default parameters, the correlation between its grants and the optimal grants stays between 0.13 and 0.18 across rounds, and its output falls short of the complete-information benchmark by 17.2 percent of no-funding output.

What the record cannot supply, then, is a signal of capability separate from resources. That is what peer review can provide under the right conditions. §6 measures what such a signal is worth, and states the conditions.

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